

Saber Ayatollahi, CFA, MSc

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SUMMARY

Senior Manager, Treasury Analytics candidate with ~7 years in Asset-Liability Management (ALM), cash-flow modelling, and quantitative interest-rate risk modelling across institutional treasury and balance-sheet mandates. Holds delegated sign-off authority at Moody's Analytics on valuation, sensitivity, and ALM outputs for multi-asset portfolios of \$5-25bn per engagement, and led design of a multi-scenario cash-flow projection engine embedding behavioural and prepayment assumptions. Reviews interest-rate derivatives valuation, yield-curve construction, and spread calibration, with hands-on stochastic/Monte Carlo and VaR/CVaR modelling from ALM advisory work. Python and SQL practitioner; CFA and dual MSc.

CORE SKILLS

Asset-Liability Management (ALM) & Balance Sheet Modelling • Cash-Flow Modelling & Liquidity Gap Analytics • Interest Rate Risk Modelling, Duration & Key-Rate Analytics • Behavioural Cash-Flow & Prepayment Assumptions • Interest Rate Derivatives Valuation (Rates, FX, Inflation) • Yield Curve Construction & Spread Calibration • Stochastic Modelling & Monte Carlo Simulation • Hedging Analysis (Interest Rate, Inflation, Currency) • Model Review, Validation, Governance & Documentation • Risk Decomposition, Attribution & Stress Testing • Python, SQL, R, MATLAB, Advanced Excel • Executive Reporting on Risk Drivers & Sensitivities

EXPERIENCE

Moody's Analytics

Asst. Director - Modelling Services (Investment & ALM Risk Lead)

Toronto, May 2023 - Present

Cash-Flow Modelling & Behavioural Risk Analytics

- Led design and implementation of an enterprise multi-asset cash-flow projection engine supporting base, stress, and reverse-stress scenarios, delivering forward-looking liquidity visibility across ALM horizons.
- Embedded behavioural cash-flow assumptions, prepayment logic, and macro stress overlays into projection models, aligning outputs with regulatory liquidity expectations and executive contingency planning.
- Architected configurable time-bucketed liquidity gap analytics (T+1 through multi-year) enabling funding optimization, refinancing-risk assessment, and asset-allocation decisions.

Interest Rate & Derivatives Valuation Analytics

- Holds delegated sign-off authority on valuation, sensitivity, and ALM outputs for institutional portfolios of \$5-25bn per engagement (~\$50bn+ cumulative), independently reviewing yield curve construction, spread calibration, and cross-asset interactions before production release.
- Validated derivatives pricing outputs across rates, FX, and inflation, confirming sensitivity consistency at portfolio-level ALM aggregates.
- Oversaw Interest Rate Risk and duration analysis under parallel and non-parallel rate-shock scenarios, consistent with industry IRRBB methodology.
- Prepared analytical summaries for senior stakeholders explaining interest-rate exposure, scenario impacts, and balance-sheet sensitivities, translating quantitative output into decision-ready narratives on risk drivers.

Model Governance, Review & Production Support

- Serves on the model governance committee, contributing to methodology review, benchmarking, documentation standards, and model-performance assessment for client-delivered analytics.
- Reviewed aggregation logic converting security-level exposures into portfolio-level risk metrics feeding downstream ALM processes; escalated outputs lacking mathematical or economic defensibility.
- Re-engineered manual spreadsheet workflows into scalable Python analytics pipelines with embedded logging, validation, and auditability controls, delivered on PFaroe DB (risk & ALM) and PFaroe PM (portfolio management).

Client Service Specialist

Toronto, May 2022 - May 2023

- Led onboarding and implementation for U.S. and Canadian pension funds, asset managers, and consulting firms on the PFaroe platform; advised on asset, liability, and ALM configuration and validated model outputs post-deployment.
- Translated client investment-team requirements into scoped development requests for Product Owners and migrated all assigned accounts from a legacy Calypso environment.

Ernst & Young (EY)

Senior Consultant - IFRS & Financial Risk Transformation

Toronto, Sep 2021 - Apr 2022

- Delivered IFRS 17 and IFRS 9 transformation projects for Canadian and international insurance clients, aligning regulatory requirements with system implementation and financial-reporting frameworks.
- Advised actuarial, finance, IT, and program-management functions on standard adoption; contributed to governance documentation and regulatory-readiness processes.

Ortec Finance Canada

Consultant - Asset-Liability Management & Risk

Toronto, Feb 2019 - Sep 2021

- Built and interpreted stochastic economic scenario generators assessing Interest Rate Risk and funding volatility under base-case and stressed environments.
- Performed cash-flow, liquidity, and hedging analysis across interest-rate, inflation, and currency risk for international institutional clients.
- Delivered ALM studies analyzing funding ratios, duration mismatch, and long-term balance-sheet risk exposures; modelled actuarial liabilities alongside all major investment-product types and economic variables.
- Performed asset-only and asset-liability (surplus) portfolio optimization on VaR and CVaR using Ortec's GLASS platform, with risk decomposition and attribution to test allocation robustness, and presented findings to pension investment committees.

EDUCATION & CREDENTIALS

- Chartered Financial Analyst (CFA) - CFA Institute, 2024
- MSc, Financial Modelling - Western University, 2018
- MSc, Chemical Engineering (Full Scholarship) - Western University, 2016